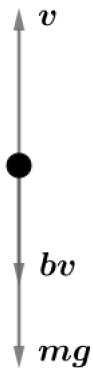


2018A F=ma Exam: Problem 1

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For an object thrown upward with air resistance, we have

$$m \frac{dv}{dt} = -bv - mg$$

$$a = -\frac{b}{m}v - g$$

The object will accelerate downwards to terminal velocity

$$0 = -\frac{b}{m}v_t - g$$

$$v_t = -\frac{mg}{b}$$

while the magnitude of acceleration decreases to 0. Thus, as $t \rightarrow \infty$, $v \rightarrow v_t < 0$. $\frac{da}{dt} = \frac{d^2v}{dt^2} > 0$ so the plot of v will be concave upwards. Hence, the answer is D.